

Creating new users: According to Microsoft, there are more than 200 million users accessing Windows all across the globe; so we can't even imagine sys admins creating accounts for all new users manually. Hence, this task is taken care of by using shell scripts.

Locking/unlocking user accounts or resetting user passwords: Just as it's difficult to create any new user account, it also becomes impossible to either lock or unlock any user's account manually, as and when required. It is not feasible to even reset any user's password manually as there may be millions of such users requesting for these activities. Therefore, all such activities are carried out with the help of various available open source tools.

Network troubleshooting: This activity requires a lot of hands-on experience and training, but open source tools (like *Wireshark*) can help ease traffic analysis with the number of powerful features that they come with.

Creating backups for Windows based systems: It becomes quite painful for administrators to create backups for Windows based systems. An open source tool named *AMANDA (Advanced Maryland Automatic Network Disk Archiver)* helps Windows sys admins set up a single master backup server, which can support both Windows desktops and other servers present over the Windows network.

Locating all devices connected to the network: A Windows sys admin must locate all the tools that connect to the network and also the various pieces of hardware that have been installed on the system. *OCS Inventory Tool* or *Zenmap* help to do this quite easily and fast.

Defragmenting different system files and registry archives: In order to ensure improvements in the performance of Windows systems, it's very important to defrag different system files, paging files and registry archives. This can be easily done using an open source tool called *UltraDefrag*.

Managing service upgrade schedules: A Windows sys admin has to manage different service upgrade schedules on a system and monitor the service uptime. An open source tool called *Nmap* is quite handy in carrying out this activity without much manual intervention.

Five famous open source tools for Windows sys admins

VirtualBox

VirtualBox is actually a must-have open source cross-platform virtualisation application for any Windows sys admin seeking to run guest open source environments on Windows, Solaris Linux or Macintosh machines. Its cross-platform property helps admins to install it on existing Intel or even AMD based computers, irrespective of whether they are running Mac, Linux, Windows or Solaris operating systems. It also extends the capabilities of the existing computer so that it can run multiple operating systems (in different virtual machines) at the same time.



Figure 1: Oracle VirtualBox manager window

Admins can run different virtual instances of various operating systems, including Windows, OpenSolaris, Linux, OpenBSD, and even DOS. VirtualBox is the only solution that is available free of cost as an open source software (OSS) under the terms of the GNU General Public License (GPL) ver2. It is a simple but powerful tool, which can run on everything, ranging from small embedded systems to different desktop machines and even cloud environments. VirtualBox is useful because of the following features:

1. It allows Windows sys admins to run more than one operating system simultaneously; hence, they can run a software written for one operating system on another one without any reboot to use it. Since they can configure the specific kind of 'virtual' hardware that should be presented to each of these operating systems, they can even install an old operating system like DOS on any computer, in spite of the fact that the computer's hardware is no longer supported by that operating system.
2. Different software vendors can use virtual machines run by VirtualBox to ship the entire configuration of a software. For example, installation of a complete mail server solution on any real machine can be a cumbersome task but VirtualBox can help to install such a complex set-up quite easily into a virtual machine.
3. Virtualisation can also reduce the hardware and electricity costs to a great extent. Computers today use only a fraction of their potential power most of the time and they run with low average system loads, wasting electricity in the process. Hence, instead of running so many physical computers that are used only partially, many virtual machines can be run on a few powerful hosts, which will balance the loads between them.
4. VirtualBox is also very useful in testing specific software with the help of virtual machines that can be run using it. It also helps in data recovery and backup.

The Performance Analysis of Logs (PAL) tool

This is a powerful tool, which reads the performance monitor counter logs, and analyses them with the help of known